

EXAM 7 – SPRING 2012

17. (4 points)

An insurer writes property and workers compensation insurance in California. To model the total annual losses for this insurer, the following two copulas are being considered:

i. $C(u,v) = uv$

ii. $C(u,v) = \min(u,v)^{0.25}(uv)^{0.75}$

a. (1 point)

Define and briefly explain the right-tail concentration function that may be used to characterize a copula.

b. (2 points)

Derive and graph the right-tail concentration functions for the two copulas.

c. (1 point)

Explain which copula is more appropriate to model total losses for this insurer. Briefly describe how the graph in part b. above supports your answer.

CONTINUED ON NEXT PAGE

Question 17 Sample Answer

Solution 1

a)

$$\begin{aligned}\text{Right tail concentration factor} = R(z) &= \frac{\Pr(X > z \text{ and } Y > z)}{1 - z} \\ &= \frac{1 + 2z + C(z, z)}{1 - z}\end{aligned}$$

The right tail concentration factor is used to look for right tail dependencies in loss distributions. Copulas such as the Gumbel and heavy-right-tail (HRT) have high values of $R\left(R = \lim_{z \rightarrow 1} R(z)\right)$ while symmetric distributions like the normal and Frank's have $R = 0$.

Solution 2

a) Right-tail concentration function

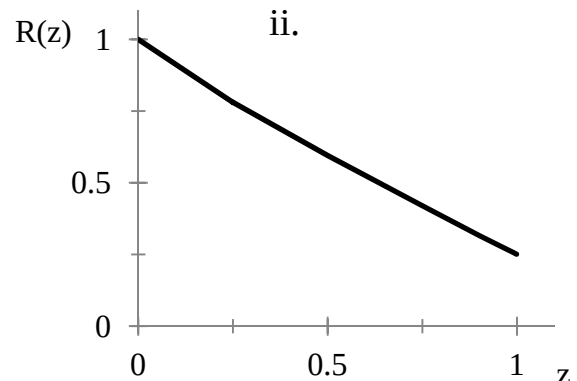
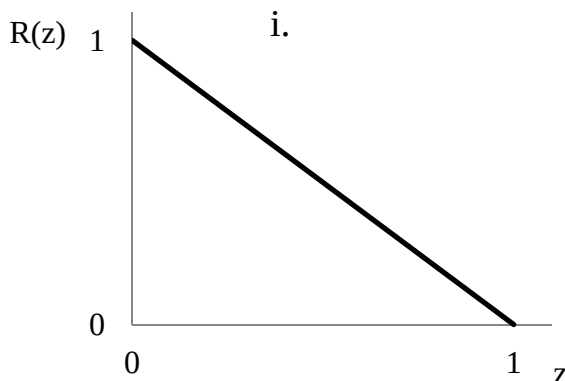
$$R(z) = \text{Prob}(U > z \text{ and } V > z) = \frac{1 - 2z + C(z, z)}{1 - z}$$

The right tail concentration function calculation how much density a copula has in its right tail. If it is large, the copula can be used to model distributions with strong correlation in the right tail.

Solution 1

b)

i.
$$R(z) = \frac{1 - 2z + z^2}{1 - z} = 1 - z$$



ii.

$$R(z) = \frac{1 - 2z + \min(z, z)^{.25} (zz)^{.75}}{1 - z} = \frac{1 - 2z + z^{.25} (z^2)^{.75}}{1 - z}$$

$$= \frac{1 - 2z + z^{1.75}}{1 - z}$$

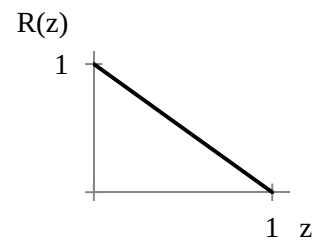
$\underline{z}$	$\underline{R(z)}$	$\underline{z}$	$\underline{R(z)}$
0	1	.9	.316
.25	.78	.95	.283
.5	.595	.99	.257
.75	.42	.999	.251

Solution 2

b)

i) $c(u, v) = uv$

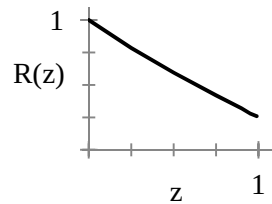
$$R(z) = \frac{1 - 2z + z^2}{1 - z} = \frac{(1 - z)^2}{1 - z} = \boxed{1 - z}$$



ii) $c(u, v) = \min(u, v)^{0.25} (uv)^{.75}$

$$R(z) = \frac{1 - 2z + (z^{.25} z^{1.5})}{1 - z} = \boxed{\frac{1 - 2z + z^{1.75}}{1 - z}}$$

$\underline{z}$	$\underline{R(z)}$
0	1
.25	.785
.5	.595
.75	.42
.9	.32
.95	.28
.99	.26



Solution 1

- c) Property and worker's compensation in California might become highly correlated in the event of a large earthquake. Therefore we need a copula with a heavy right tail. The second copula has a heavier right-tail. The second copula has a heavier right-tail. Therefore it would be more appropriate than the first (the product copula)

Solution 2

- c) WC & property are generally independent. However, there may be correlations in extreme events. For example, an earthquake during a work day could cause property damage and injuries to employees. Thus $R(z) > 0$ as $z \rightarrow 1$ could be appropriate. For copula ii, $R(z) \rightarrow .25$ as $z \rightarrow 1$ but for copula i, $R(z) \rightarrow 0$. Thus copula ii is more appropriate (the two are similar for smaller z , which also what we'd expect: normally independent)

Examiner Comment

For the a. part, many candidates offered a definition for right-tail concentration function or the right-tail concentration formula itself. It was common for candidates not to have both the written explanation and the formula.

Most candidates had difficulty with the b. part. Candidates often failed to both derive the function and graph the function as required by the question. When graphing the functions, Candidates commonly failed to label their graphs or show any work as to how they plotted the points on the graph. Additionally, it was common for candidates on the second copula to intercept $R(1)$ with the x-axis, this was incorrect.

For the c. part, many candidates correctly selected the proper copula but failed to explain why that copula was the best choice, and in some cases there were no graphs or data present in part b. but the candidate was still able to guess and select correctly. We required a visual graph to be present in b. in order to get full credit in c. We also gave limited partial credit to candidates who selected the first copula as the answer and provided an explanation that property and workers compensation losses are independent.